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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION – MARCH 2026
FOURTH SEMESTER
Discipline Specific Core Course (DSC) -MG4DSCZGY200, MG4DSCZIM200 -
ANIMAL DIVERSITY CHORDATA I

Answer Scheme

Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I), and Appreciation (Ap).

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer all questions. Each question carries 1 mark.

1. Warm-blooded vertebrates include: **[U]/[CO1]**

Ans**: b) Aves and Mammals**

2. Example of super order Chondrostei is: **[U]/[CO1]**

Ans: **c) Acipenser**

3. How many lungs is functional in snakes **[U]/[CO1]**

Ans: **a) One**

4. What characteristic makes chordates deuterostomes? **[U]/[CO1]**

Ans: **c) Anus is formed near the blastopore**

5. The migration of hilsa fish is **[U]/[CO3]**

Ans: **a) Anadromous**

6. Which duct of the frog is associated with the liver? **[U]/[CO2]**

Ans: **b) Hepatic duct**

7. Which siphon brings water into the body of Ascidia? **[K]/[CO1]**

Ans: **c) Inhalant siphon**

8. Accessory respiration is common in fishes of: **[U]/[CO3]**

Ans: **b) Stagnant Ponds**

9. Fertilization in *Euphlyctis hexadactylus* occurs in: **[K]/[CO2]**

Ans: **c) Water**

10. A common example of Cyclostomata is **[U]/[CO1]**

Ans: **d) Petromyzon**

11. The heart of fishes pumps **[U]/[CO1]**

Ans: **b) Only deoxygenated blood**

12. Which adaptation of the hind limbs enables frogs to swim efficiently? **[U]/[CO2]**

Ans: b**) Webbing between toes**

Part B – Short Answer Questions

*(*Answer any **6** questions – Each question carries **3** marks*)*

13. **Mention two neotenic features of Larvacea. [U]/[CO1]**
    Ans: Tail retained in adult; Notochord persists throughout life; Dorsal nerve cord remains present; Free swimming habit **(Any two features: 3 marks**)

14. **What are lungfishes? [U]/[CO1]**
    Ans: Lung fishes are a group of fishes belonging to the order Dipnoi that possess lungs in addition to gills for respiration. Found in freshwater habitats; Capable of aestivation (dormancy) during dry conditions. Examples: *Protopterus*, *Lepidosiren*, and *Neoceratodus*. (**Definition: 1 mark; Feature: 1 mark; Example: 1 mark**)

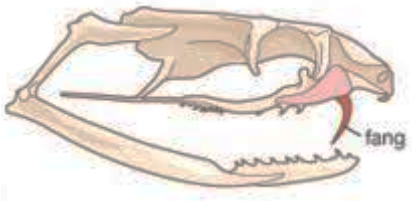
15. **Differentiate between cycloid and ganoid scales. [A]/[CO3]**
    Ans: Cycloid Scales: Thin, round and flexible; smooth edged; Growth rings seen clearly; Found in modern bony fishes
    Ganoid Scales: Thick, hard and rhomboidal shaped; Covered with shiny enamel-like **ganoine**; Found in primitive fishes (e.g., *Lepisosteus*, *Polypterus*) (**Structure and shape: 2 marks; Example: 1 mark**)

16. **What are the salient features of Chelone? [U]/[CO1]**
    Body covered with a hard shell consist of dorsal carapace and ventral plastron; Toothless jaws with horny beak; Limbs are paddle like; Small head, short neck and tail is rudimentary; Oviparous (lay eggs on land); Respiration by lungs. (**Any 3 points: 3 marks**)

17. **Mention the divisions of the nervous system in a frog. [K]/[CO2]**
    Ans: The nervous system of a frog is divided into three parts: 1) Central Nervous System (CNS): Consists of brain and spinal cord 2) Peripheral Nervous System (PNS): Includes cranial and spinal nerves; Autonomic Nervous System (ANS): Controls involuntary activities like heartbeat and digestion. (**1 mark each**)

18. **Describe proteroglyphous fangs with a diagram. [K]/[CO3]**

Ans: Proteroglyphous fangs are short, fixed, poison fangs located at the front of the upper jaw in certain venomous snakes.



Key features: Fangs are small, hollow or grooved for venom conduction;
Fixed (non-movable) and always erect;
Venom is injected directly into the prey;
Found in snakes like the Cobra and krait.
(**Structure: 1 mark; Any 1 feature: 1 mark; Diagram: 1 mark**) (***Liberal Valuation***)

19. **Explain the life cycle of Thaliacea. [U]/[CO1]**

Ans: Thaliaceans show alternation of generations involving asexual and sexual phases. The life cycle includes two forms: oozooid (asexual form) and blastozooid (sexual form). The oozooid reproduces asexually by budding and produces blastozooids. The blastozooids are sexual forms and produce gametes. After fertilization, a zygote develops into a new oozooid, completing the cycle. (**Mention the term Alternation of generation: 1 mark; Explanation: 2 marks**)

20. **Describe the Structure and function of the calcaneum in frogs. [U]/[CO2]**

Ans: Calcaneum is a long, elongated bone present in the hind limb of the frog. It is one of the tarsal bones, lying parallel to the astragalus. **Function:** Helps in jumping and locomotion, by providing leverage to the hind limbs; Provides support and flexibility to the ankle joint. (**Structure: 2 mark; Any 1 function: 1 mark**) (***Liberal valuation***)

Part C

Short Essay Questions

Answer any 4 questions. Each question carries 5 marks.

21. **Compare the anatomy of Cephalochordata with vertebrates. [U]/[CO1]**

1) In cephalochordates, the notochord persists throughout life and extends along the entire body length, whereas in vertebrates it is present only in the embryonic stage and is later replaced by a vertebral column**.** 2) Cephalochordates lack a vertebral column, while vertebrates possess a well-developed backbone. 3) In cephalochordates, the head is not distinct and the brain is simple and poorly developed; vertebrates have a well-defined head with a highly developed brain. 4) The sense organs in cephalochordates are poorly developed, whereas vertebrates have well-developed and specialized sense organs. 5) Cephalochordates have a closed circulatory system without a true heart, while vertebrates possess a closed circulatory system with a well-developed heart**. (Any 5 points: 5 marks)**

22. **Describe the major morphological characters used in the subclassification of Chondrichthyes. [U]/[CO1]**

Ans: Subclass Chondrichthyes is divided into Elasmobranchii (sharks and rays) and Holocephali (chimaeras) based on the following morphological features: **Gill openings:**

Elasmobranchii: 5–7 pairs of separate gill slits, no operculum; Holocephali: Gill slits covered by an operculum **Position of mouth:** Elasmobranchii: Ventral mouth ; Holocephali: Terminal mouth **Type of scales:** Elasmobranchii: Placoid scales present ; Holocephali: Scales absent or reduced **Spiracle:** Elasmobranchii: Spiracle usually present ;Holocephali: Spiracle absent **Tail :** Elasmobranchii: Heterocercal tail ; Holocephali: Diphycercal tail (**Any 5 points: 5 marks**)

23. **Describe parental care in fishes. [U]/[CO3]**
Ans: Parental care in fishes refers to the protection and care of eggs and young ones to increase their chances of survival. **Types of parental care- Substrate Brooding:** Females deposit eggs on substrate and guard them until hatching**; Nest building:** Some fishes build nests to lay eggs. Example: Stickleback prepares a nest and guards it. **Mouth brooding:** Eggs are kept in the mouth of the parent for protection. Example: Tilapia. **Brood pouch:** In Hippocampus, eggs are kept in a brood pouch of the male until they develop. **Significance:** Increases survival rate of offspring; Provides protection from predators and environmental hazards (**Definition: 1 mark; Explanation- Any 3 types: 3 marks; Significance: 1 mark)**

24. **Give the Salient features of order Urodela. Substantiate it with an example. [K]/[CO1]**
Lizard like elongated body with a distinct head, trunk and tail; Tail persists throughout life; Two pairs of limbs present; Skin is moist, glandular and without scales; Tympanum is absent; Internal Fertilization; Example**:** Salamander (*Amblystoma*) (**Any Four Features: 4 marks; Example: 1 mark**)

25. **Differentiate between the systemic and pulmocutaneous arteries in frog, mentioning their respective functions. [U]/[CO2]**
Ans: **Systemic Arteries:** Arise from the truncus arteriosus as a pair of systemic arches; Carry oxygenated blood from the heart to all parts of the body (except lungs and skin); Supply organs like head, trunk and limbs; Function: Distribution of oxygenated blood to body tissues. **Pulmocutaneous Arteries:** Arise from the truncus arteriosus as pulmocutaneous arches; Carry deoxygenated blood to lungs and skin; Involved in both pulmonary (lungs) and cutaneous (skin) respiration; Function: Oxygenation of blood through lungs and skin. (**Any 3 distinguishing points: 3 marks; Function: 2 marks**)

26. **Distinguish between *Eryx johnii* and *Eryx conicus*. [U]/[CO3]**
Ans*: Eryx johnii* is a non-venomous burrowing snake with a uniform reddish-brown body and a very blunt rounded tail that resembles the head, giving a false “two-headed” appearance. Its scales are comparatively smoother and it lives in loose soil where it feeds on small rodents.
Eryx conicus is also non-venomous but differs in having brown coloration with dark blotches and rougher keeled scales. Its tail is less rounded than that of *E. johnii*, and it inhabits sandy or dry areas. (**Any 5 distinguishing features: 5 marks**)